Six label-free probes of Laya: order and position sensitivity, a wording cliff, a Tagalog gap, an option-count collapse, and an escalation head that never fires

PinoAI research note · THEMIS-BENCH-003 · THEMIS-PROBES v0.1, both routings · v1.1

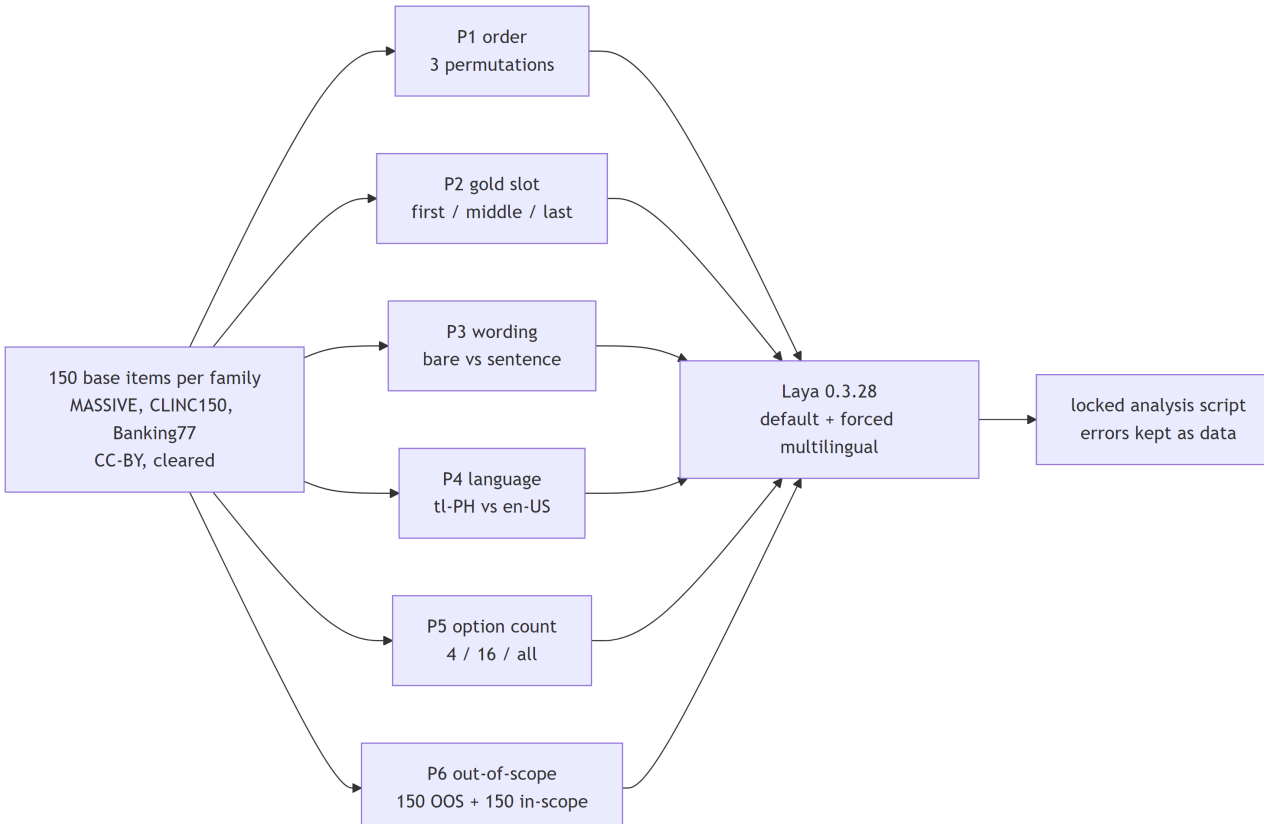
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Abstract

THEMIS-BENCH-001 and –002 characterised Laya on one benchmark each. This note widens the lens: six families of label-free probes, each built so that a robust judge should give the same answer, or an answer that degrades only gracefully, when one thing changes - the order of the options, the slot of the correct option, the wording of the options, the language of the request, the number of options, or whether the request is answerable at all. Laya, the open comparator Themis must beat, was run on all 2,250 probes under both of its routings and analysed by a script locked before the runs landed. The results, below, are the concrete robustness targets Themis is designed against.

1. The probes and the runs



Six probe families, one base item each, one controlled change per family; metrics declared before any judge ran.

THEMIS-PROBES v0.1 (data-science) holds six families of 150 base items each, 2,250 rows in all, drawn from the same cleared CC-BY sources as the Themis item pool and loaded through the pool builder's own asserts; no probe item appears

in the pool. Every variant asserted its own effect before it was written (the PM-023 rule), and a mutation - moving a gold option to the wrong slot - fails the build. The tl items in P4 are MASSIVE's human translations into Tagalog, not native or Taglish text.

ml-engineer ran Laya 0.3.28 at the registered pins on every probe, CPU, offline, under both routings: the default router and forced multilingual. Errors are kept as data, never retried. The analysis script was registered before the full runs landed; paired comparisons use exact McNemar tests on complete base items only, and the number of items dropped for an error in any variant is printed.

Scope of the default run. On the default route every probe built on CLINC150′s 150 intents errors, because the English checkpoint's 512-token window cannot place 150 option markers. P1, P2, P3 and P5 on that route are therefore read on the 113 complete non-CLINC base items, and P6 (CLINC only) is not measurable there. The forced-multilingual route (1,024-token window) ran every row without error.

2. Results

Table 1: THEMIS-PROBES v0.1 on Laya 0.3.28, one run per route. Descriptive; no family-wise claim.

Family	What changes	Default route	Forced multilingual
P1 order	same options, 3 orders	same answer 23/113 = 0.20	33/150 = 0.22
P2 position	gold first / middle / last	0.17 / 0.20 / 0.40; last vs first 29 vs 3, p = 3e-06	0.20 / 0.29 / 0.19; middle vs last 28 vs 12, p = 0.017; first vs last p = 0.83
P3 wording	bare name vs "This message is about X."	23 -> 1 of 113	33 -> 2 of 150; same choice 0/150
P4 language	same utterance, tl vs en	tl 17 vs en 31 of 150; 6 vs 20 discordant, p = 0.009	tl 16 vs en 27 of 150; 5 vs 16, p = 0.027
P5 options	4 / 16 / all options	82 / 66 / 23 of 113	115 / 100 / 34 of 150; drop 0.54 [0.45, 0.63]
P6 out-of-scope	answerable vs not	not measurable (150-option window)	escalation rate 0/150 OOS, 0/150 in-scope; act-probability AUROC 0.50 (one distinct value); 1 - confidence AUROC 0.739 [0.68, 0.79]; in-scope accuracy 60/150

3. Reading

Order and position. Shuffling the same options changes Laya's answer four times in five. Its accuracy also depends on where the correct option sits, but not in one direction: on the default route the last slot is favoured, on the multilingual route the middle, and in BENCH-002′s sentence-style options the collapse fell on the first. The stable finding is position sensitivity, whose direction depends on the route and the option style - not a fixed slot bias that could be corrected with a constant.

Wording. Wrapping the same category names in a short sentence ("This message is about X.") takes accuracy to about one percent on both routes, and on the multilingual route not a single item keeps its answer. This replicates the BENCH-001/002 wording effect on new items and a second source; it is the largest single effect in the probes.

Language. The same request, translated into Tagalog by humans, scores lower than in English on both routes - 9.3 points on the default route, 7.3 on the forced multilingual route - and the two languages agree on only 15-32% of items. On the default route most Tagalog probes (124/150) and all English probes (150/150) are read by the same English checkpoint. The gap is similar on both routes (9.3 vs 7.3 points); forcing the multilingual checkpoint does not remove it, so routing alone does not explain it. No route-by-language test was run, and the cause is not claimed.

Option count. Accuracy falls by about half from four options to all of them, and on the English route a 150-option question cannot be posed at all. Laya's option capacity is a hard architectural ceiling, not only a soft accuracy cost.

Escalation. Laya as released never escalates: its act probability takes a single value on every probe, so the escalation signal is degenerate and its AUROC is 0.5 by ties. Yet the information is there - Laya's own confidence separates out-of-scope from in-scope requests with AUROC 0.739. The signal exists; it is simply not wired to the act decision.

4. What this changes for Themis

- **A fair abstention baseline** (registered before any Themis score): Themis's escalation is compared with "Laya plus a threshold on 1 - confidence", cross-fitted on the Themis v1 DEV split by twin engine D and read on TEST once, with

the same procedure applied to Themis's own confidence. Beating Laya's dead act head would be a wiring win and is not reported as a model win. Design target W2 (AUROC >= 0.85) is now read against 0.739, not the README's 0.30.
- **Robustness targets as evaluations:** order consistency (P1) and label-swap consistency (W5) at >= 0.99; no position dependence (P2); wording robustness (P3); Tagalog within three points of English on the same utterance, on any route (P4, W8); many-option accuracy with no hard option-count ceiling (P5, W1).
- **The comparator configuration:** Laya at its best route per question type, recorded, on identical items.

5. Limitations

- One run per route, n = 113-150 per family; descriptive, with per-family tests and no family-wise correction.
- On the default route the 150-option items error, so its P1/P2/P3/P5 rows rest on 113 non-CLINC base items and its P6 is not measurable.
- The tl probes are human translations of an English dataset, not native Tagalog or Taglish; v0.2 adds human-written Taglish items reviewed by a native speaker.
- The probes have no train/dev split, so no threshold or calibration is fitted on them.

Reproducibility

Table 2: Files.

Item	Where
Probes	`data/eval/themis/probes/themis-probes-v0.1/probes.jsonl` (sha256 38ac7f01), plan `experiments/plans/THEMIS-PROBES-v0.1.md`
Runs	`experiments/results/themis/laya-probes/preds-full-{default,multilingual}.jsonl` (78157a39, 0c12a317; git-ignored)
Analysis	`scripts/eval/analyze_themis_probes.py` (8634e52e, registered before the runs landed) -> `analysis-full-*.json`
Numbers in this note	`experiments/reports/_themis_bench003.py` asserts each one

Acknowledgments

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